SELF-HEALING INFRASTRUCTURE WITH AUTO SCALING USING AWS

Project Title

Self-Healing Infrastructure with Auto Scaling using AWS

Abstract

Cloud computing provides scalable, reliable, and cost-effective infrastructure for modern applications. However, traditional systems depend on manual monitoring to detect failures and handle changing workloads, resulting in downtime and inefficient resource utilization. This project implements a Self-Healing Infrastructure with Auto Scaling using Amazon Web Services (AWS). It combines Amazon EC2, Elastic Load Balancer (ELB), Auto Scaling Groups (ASG), Amazon CloudWatch, AWS Identity and Access Management (IAM), and Amazon Virtual Private Cloud (VPC) to create an automated cloud environment. CloudWatch continuously monitors the health and performance of EC2 instances and triggers Auto Scaling policies whenever CPU utilization or health status crosses predefined thresholds. The Auto Scaling Group automatically launches new instances during increased demand, terminates unnecessary instances during low traffic, and replaces failed instances without human intervention. The Elastic Load Balancer distributes incoming requests among healthy servers to ensure uninterrupted application availability. The project aims to improve reliability, fault tolerance, scalability, and cost efficiency while minimizing administrative effort. The expected outcome is a resilient cloud infrastructure capable of maintaining continuous service, reducing downtime, and optimizing cloud resources through automation.

Introduction

Modern businesses require applications that remain available at all times despite hardware failures or sudden increases in user traffic. Maintaining such systems manually is expensive and time-consuming. AWS provides services that automate infrastructure management, allowing organizations to build highly available and scalable systems. In this project, CloudWatch monitors application health, Auto Scaling adjusts the number of EC2 instances based on workload, and Elastic Load Balancer distributes requests across healthy instances. This automated approach ensures continuous service, improves performance, reduces operational costs, and provides practical exposure to real-world cloud architecture.

Key Points of the Project

- Automatic health monitoring using Amazon CloudWatch.
- Automatic replacement of failed EC2 instances.
- Dynamic scaling based on workload demand.
- Load balancing using Elastic Load Balancer.
- High availability and fault tolerance.
- Secure access using AWS IAM.
- Cost optimization through automated resource management.

Advantages & Limitations

Advantages

- High availability
- Reduced downtime
- Automatic recovery
- Improved scalability
- Better resource utilization
- Lower operational cost

Limitations

- Requires AWS account
- Cloud usage charges
- Complex initial setup
- Requires internet connectivity

Components in the System Architecture

User: Sends requests to the cloud application.

Elastic Load Balancer: Distributes requests among healthy EC2 instances.

Amazon EC2: Hosts and runs the application.

Auto Scaling Group: Adds, removes, or replaces instances automatically.

Amazon CloudWatch: Monitors resources and triggers alarms.

AWS IAM: Provides secure authentication and authorization.

Amazon VPC: Offers secure network isolation.

Future Scope

- Predictive auto scaling using AI.
- Integration with AWS Lambda.
- Container deployment using Amazon EKS.
- Database high availability using Amazon RDS Multi-AZ.

Conclusion

The Self-Healing Infrastructure with Auto Scaling using AWS demonstrates how cloud automation improves application availability, reliability, and scalability. By integrating EC2, Auto Scaling, Elastic Load Balancer, CloudWatch, IAM, and VPC, the infrastructure can monitor itself, recover from failures, and efficiently manage workloads with minimal human intervention. This project provides practical knowledge of AWS cloud services while demonstrating industry-standard techniques for building resilient and cost-effective applications.